HAPTIGUIDE AI / GARMENT ENVELOPE / ORTHOGRAPHICS

Simplified faceted textile shell: no mannequins or human representations. Projection is diagrammatic.

FRONT ELEVATION



REAR ELEVATION



MOTOR POSITION REFERENCE: FRONT is Y-negative / REAR is Y-positive

TORSO DESIGN STATIONS - RADIAL ENVELOPE

Z from waist	Half width X	Half depth Y
0	218	117
85	220	120
280	236	127
365	255	133
408	250	123
455	98	71
487	97	70

NOMINAL HAPTIC CENTERS (GLOBAL X / GLOBAL Z)

L shoulder:	X = +141	Z = 389
R shoulder:	X = -141	Z = 389
Upper-back:	X = 0	Z = 301
Lower-back:	X = 0	Z = 115

Y is normal-offset from local back contour. See JSON for full XYZ.

DWG HG-AI-CONCEPT-01

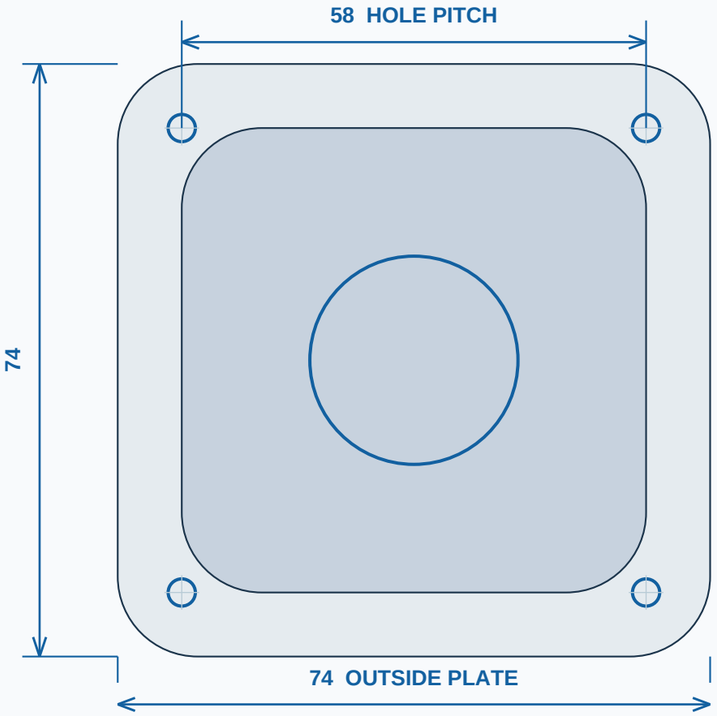
UNIT mm | REV A | ADULT-MEDIUM DESIGN ENVELOPE

Conceptual only. Not checked for actual hardware, human fit, comfort, water ingress or safety.

Not a release-to-manufacture print

Nominal machining references only; motor/board/cell/hardware envelopes must be measured before fabrication.

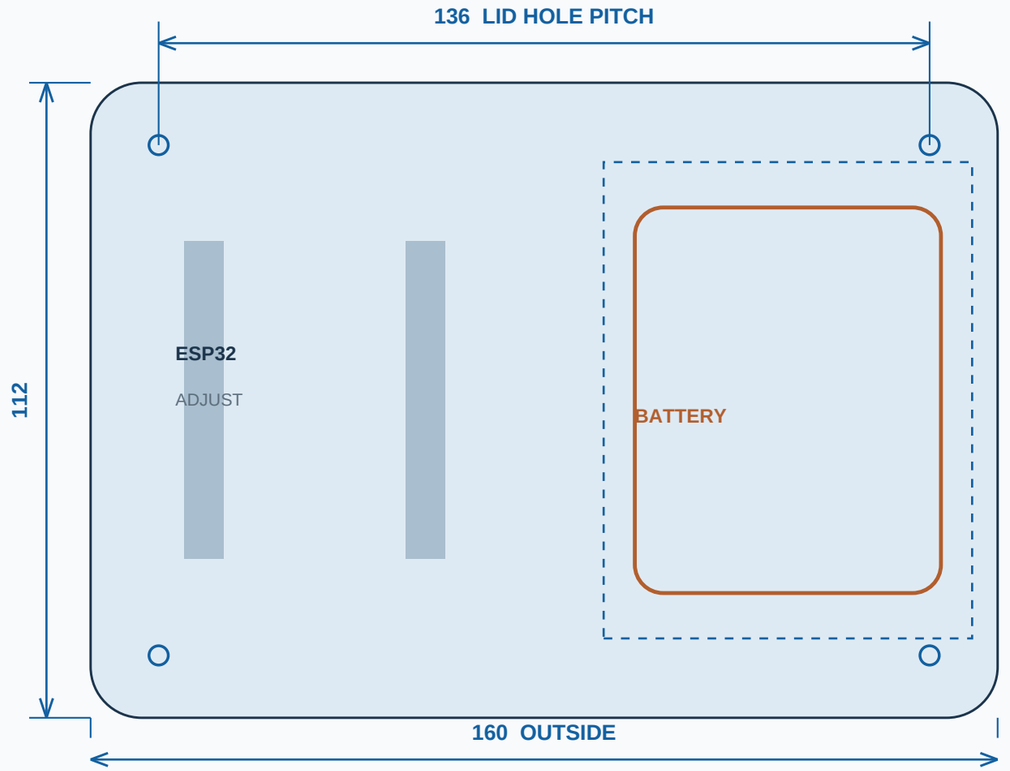
MOTOR MODULE MOUNT / TOP



4X DIAMETER 3.4 mm clearance for M3

Motor-cavity diameter 26; depth 10 (illustrative only).

ELECTRONICS LID / TOP



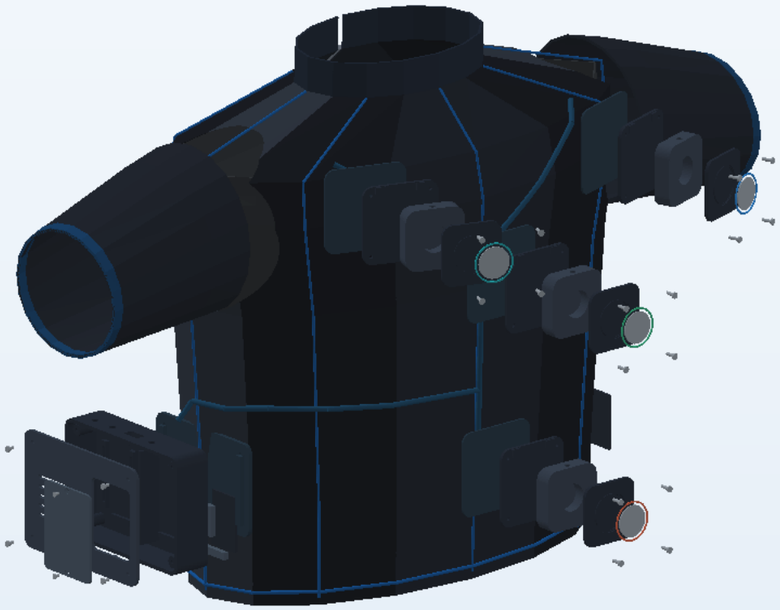
4X DIAMETER 3.4 lid holes at local X = +/-68, Y = +/-45.

Battery hatch: 54 x 68, centered at local X = +43, Y = 0.

ADDITIONAL SECTION / INTERFACE DIMENSIONS (mm)

Motor plate thickness:	**3.5**	Electronics lid thickness:	**3.2**
Motor enclosure height:	**16.0**	Battery access door:	**65 x 84 x 2.8**
Motor cap thickness:	**3.2**	Dock plate:	**132 x 88 x 4.3**
Outer electronics depth:	**37.0**	Garment shell nominal:	**2.4**
Enclosure wall / floor:	**3.5 / 3.2**	Sleeve cuff X:	**+/- 385**

Exploded offsets are illustrative. They are not validated assembly travel or mechanisms.



QUICK REMOVAL / MAINTENANCE

1. Unclip nylon side buckle tabs.
2. Open the full-length front zipper.
3. Release the side electronics latch.
4. Slide electronics off its waist dock.
5. Remove motor cover / plate screws.
6. Replace fabric carrier separately.

RESERVED HARDWARE ZONES

ESP32 module: adjustable rails only

Rechargeable battery: compartment only

Regulator + driver: blank mount areas

USB / switch / wire exits: generic

Heat-set inserts: not specified

REQUIRED PRE-PROTOTYPE VERIFICATION

- Measure actual ESP32, motor, cell, regulator and driver; revise parametric clearances and mounting patterns.
- Prototype seam and armhole shape, textile stresses, buckle engagement and sliding dock/latch travel.
- Verify wearer clearance, vibration coupling, battery protection / thermal isolation, connector strain relief.
- Check rigid-body collision, screw access, ingress resistance, printing tolerances and wiring/harness service loops.